INS 401: Project Management

Managing Project Scheduling

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24 November 2025
Main Campus, Lagos

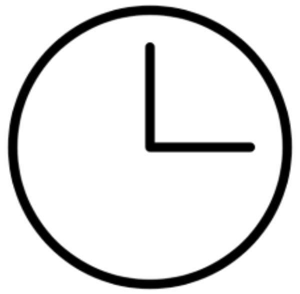
Ground Rules

- Participate

- Respect each other

- Silence Phones

- Be on time



Outline

- Project schedule management
- Common problems in project scheduling
- Tools and techniques for project scheduling

Learning Outcomes

At the end of this class, students should be able to:

- discuss project schedule management
- describe common problems in project scheduling
- discuss the tools and techniques for project scheduling

Project Schedule Management

Project schedule management includes the processes required to manage the timely completion of the project. Project scheduling provides a detailed plan that represents how and when the project will deliver the products, services and results defined in the project scope.

The processes involved in project schedule management include:

- Plan Schedule Management
- Define Activities
- Sequence Activities
- Estimate Activity Durations
- Develop Schedule
- Control Schedule

- PMI Process Groups and Knowledge Areas Mapping

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PROCESS GROUP AND KNOWLEDGE AREA MAPPING

Knowledge Areas	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Source: *PMBOK® Guide*, 6th Edition

Plan Schedule Management

Plan Schedule Management is the process of establishing the policies, procedures and documentation for planning, developing, managing and controlling the project schedule. It provides guidance and direction on how the project schedule will be managed throughout the project.

Key Output

- **Schedule management plan** – a component of the project management plan that establishes the criteria and the activities for developing, monitoring and controlling the schedule.

Plan Schedule Management

Inputs	Tools & Techniques	Outputs
1. Project charter 2. Project management plan ▪ Scope management plan ▪ Development approach 3. Enterprise environmental factors 4. Organizational process assets	1. Expert judgement 2. Data analysis -Alternative analysis 3. Meetings	1. Schedule management plan

Define Activities

Define activities is the process of identifying and documenting the specific actions to be performed to produce the project deliverables. It decomposes work packages into schedule activities that provide the basis for estimating, scheduling, executing, monitoring and controlling the project work.

Key Outputs

- **Activity list** – includes the schedule activities required on the project. It includes an activity identifier and a scope of work description for each activity.
- **Activity attributes –** these attributes extend the description of the activity by identifying multiple components associated with each activity. The details may include the unique activity ID, WBS ID, an activity label or name, predecessor or successor relationships, etc.
- **Milestone list** – a milestone is a significant point or event in a project. Milestones have zero duration since they represent a significant point or event.

Define Activities

Inputs

1. Project management plan
   - -Schedule management plan
   - -Scope baseline
2. Enterprise Environmental factors
3. Organizational process assets

Tools & Techniques

1. Expert judgement
2. Decomposition
3. Rolling wave planning
4. Meetings

Outputs

1. Activity list
2. Activity attributes
3. Milestone list
4. Change requests
5. Project management plan updates
   - -Schedule baseline
   - -Cost baseline

Sequence Activities

Sequence activities is the process of identifying and documenting relationships among the project activities. It defines the logical sequence of work to obtain the greatest efficiency given all project constraints.

Key Output

- **Project schedule network diagrams** – this is a graphical representation of the logical relationships (also known as ***dependencies***) among the project schedule activities. The diagram can be produced manually or by using a project management software.

Sequence Activities

Inputs	Tools & Techniques	Outputs
1. Project management plan -Schedule management plan -Scope baseline 2. Project documents -Activity attributes -Activity list -Assumption log -Milestone list 3. Enterprise Environmental factors 4. Organizational process assets	1. Precedence diagramming method 2. Dependency determination and integration 3. Leads and lags 4. Project management information system	1. Project schedule network diagrams 2. Project documents updates -Activity attributes -Activity list -Assumption log -Milestone list

Precedence Diagramming Method Relationship Types

Estimate Activity Durations

Estimate activity durations is the process of estimating the number of work periods needed to complete individual activities with estimated resources. It provides the amount of time each activity will take to complete.

Key Outputs

- **Duration estimates** – these are quantitative assessments of the likely number of time periods that are required to complete an activity.
- **Basis of estimates** – a supporting documentation that provides a clear understanding of how the duration estimate was derived.

Estimate Activity Durations

Inputs

1. Project management plan
   - -Schedule management plan
   - -Scope baseline
2. Project documents
   - -Activity attributes
   - -Activity list
   - -Assumption log
   - -Lessons learned register
   - -Milestone list
   - -Project team assignments
   - -Resource breakdown structure
   - -Resource calendars
   - -Resource requirements
   - -Risk register
3. Enterprise Environmental factors
4. Organizational process assets

Tools & Techniques

1. Expert judgement
2. Analogous estimating
3. Parametric estimating
4. Three-point estimating
5. Bottom-up estimating
6. Data analysis
   - -Alternative analysis
   - -Reserve analysis
7. Decision making
8. Meetings

Outputs

1. Duration estimates
2. Basis of estimates
3. Project documents updates
   - -Activity attributes
   - -Assumption log
   - -Lessons learned register

Develop Schedule

Develop schedule is the process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create a schedule model for project execution and monitoring and controlling. It generates a schedule model with planned dates for completing project activities.

Key Outputs

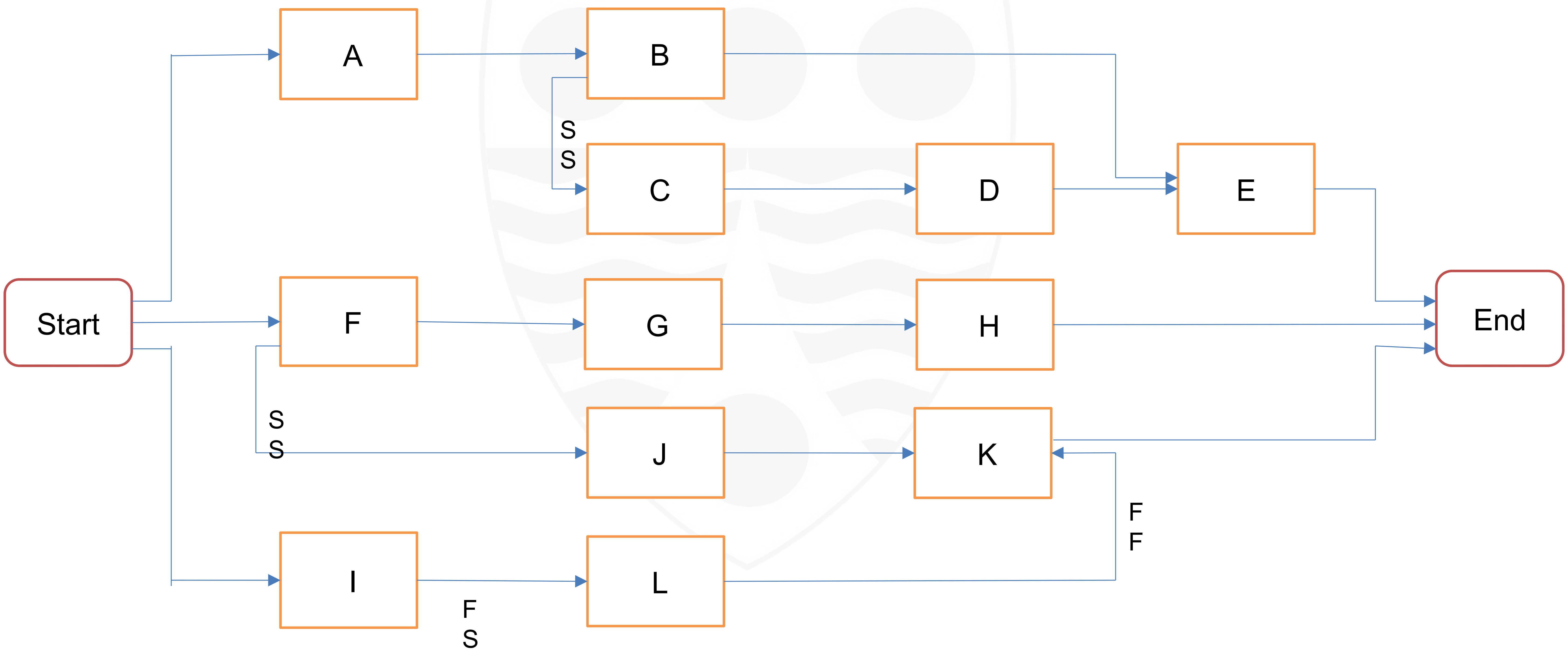
- **Schedule baseline** – this is the approved version of a schedule model that can be changed only through formal change control procedures and is used as a basis for comparison to actual results.

- **Project schedule** – this is an output of a schedule model that presents linked activities with planned dates, durations, milestones and resources.

Develop Schedule

Inputs	Tools & Techniques	Outputs
1. Project management plan -Schedule management plan -Scope baseline 2. Project documents -Activity attributes -Activity list -Assumption log -Lessons learned register -Milestone list -Project team assignments -Project schedule network diagrams -Resource calendars -Resource requirements -Risk register 3. Enterprise Environmental factors 4. Organizational process assets	1. Schedule network analysis 2. Critical path method 3. Resource optimization 4. Data analysis -What-if scenario analysis -Simulation 5. Leads and lags 6. Schedule compression 7. Project management information system 8. Agile release planning	1. Schedule baseline 2. Project schedule 3. Schedule data 4. Project calendars 5. Change request 6. Project management plan updates -Schedule management plan -Cost baseline 7. Project documents updates -Activity attributes -Assumption log -Duration estimates -Lessons learned register -Resource requirements -Risk register

Project Schedule Network Diagram



Critical Path Method [Key Points]

- The critical path method is used to estimate the minimum project duration and determine the amount of schedule flexibility on the logical network paths within the schedule model.

- The critical path is the sequence of activities that represents the longest path through a project, which determines the shortest possible project duration.

- The critical path usually has a zero total float. The total float (TF) represents the extra amount of time to complete the tasks on a non-critical path.

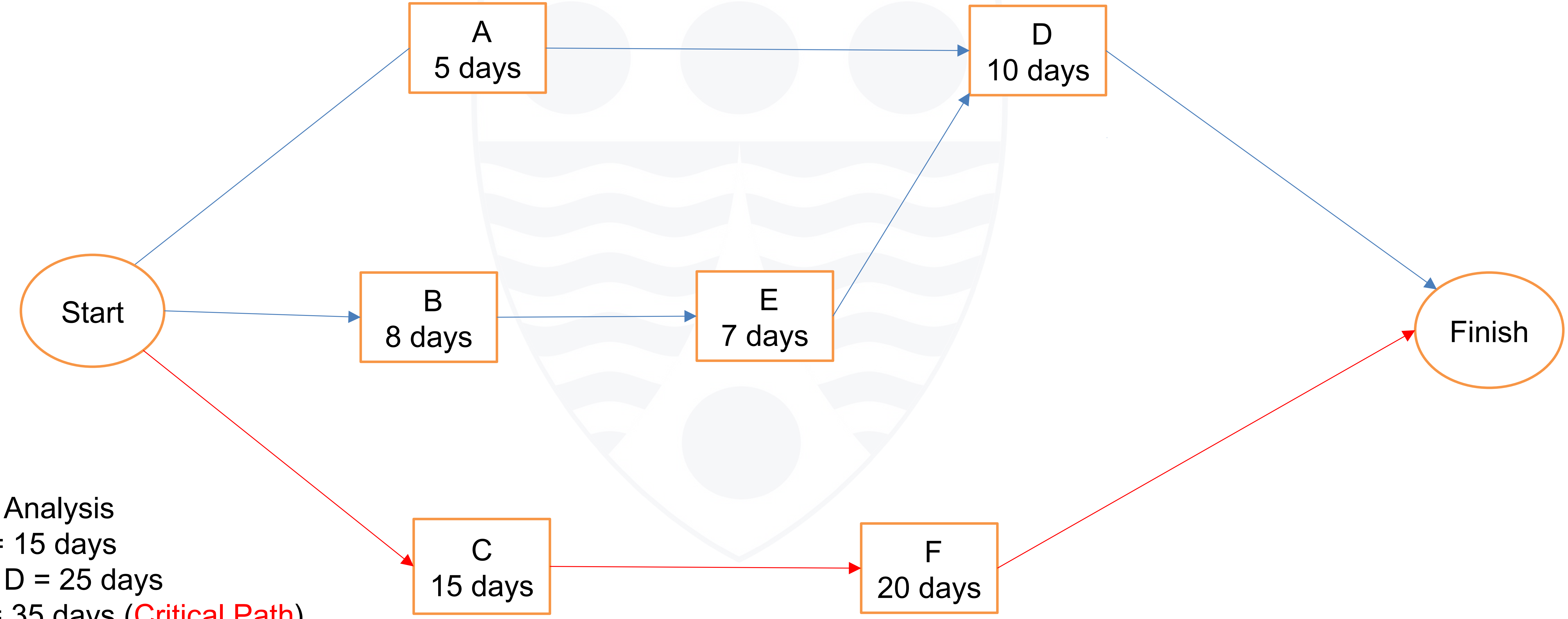
- Using the forward pass to calculate the early start (ES) and early finish (EF) of activities highlights the critical path.

- Using the backward pass to calculate the late start (LS) and late finish (LF) of activities highlights the float in the system.

- To calculate the total float; TF = LF-EF or TF= LS-ES

- You may use the Gantt Chart to create or visualize the Project Schedule. This shows the task dependencies, durations and the critical path analysis.

Critical Path Method [Example 1]



Paths Analysis
A- D = 15 days
B - E- D = 25 days
C- F = 35 days (Critical Path)

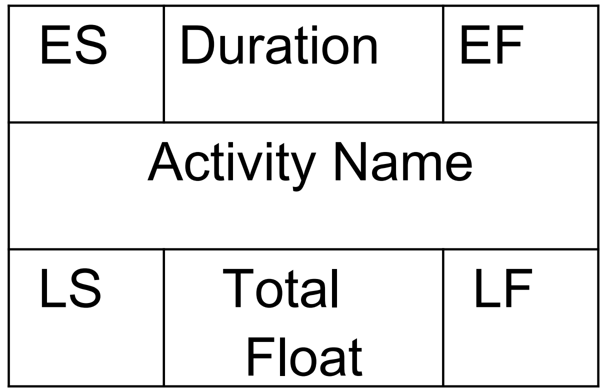
Critical Path Method [Practical Example]

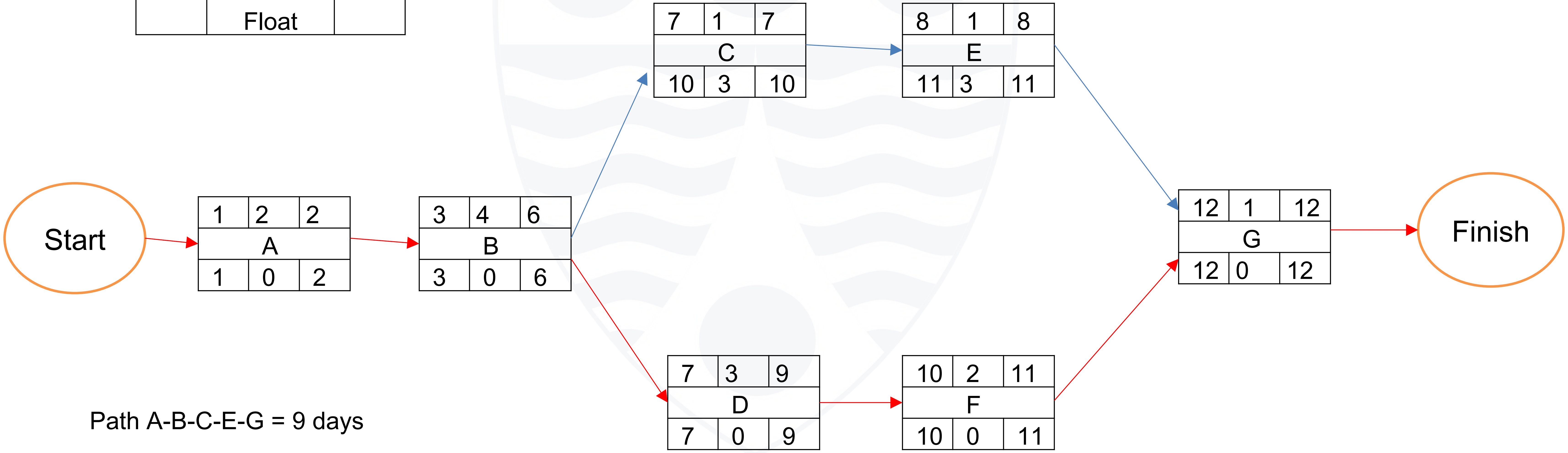
Let us consider a project to buy and install computers and solar system for a school. From the tasks and their durations given below, determine the critical path and the total float in the system.

Task Designation	Task Description	Task Duration (days)
A	Make a budget	2
B	Purchase items	4
C	Install computers	1
D	Install solar system	3
E	Test computers	1
F	Test solar system	2
G	Deployment	1

Critical Path Method [Solution to Problem]

LEGEND





Path A-B-C-E-G = 9 days

Path A-B-D-F-G = 12 days (Critical path)

Total float = 3 days

Project Evaluation and Review Technique [PERT]

- PERT is a framework that helps project managers to visualize uncertain project timelines, project tasks and task dependencies.

- PERT works by allowing project managers to identify project tasks, define three possible time estimates (optimistic –O; pessimistic – P; and most likely – M) and determine a project's critical path.

STEPS

- Define project activities or tasks
- Identify dependencies between activities
- Calculate activity or task durations using the formula

  Expected time (E) = [O+(4xM)+P]/6
- Create network diagram
- Identify the critical path

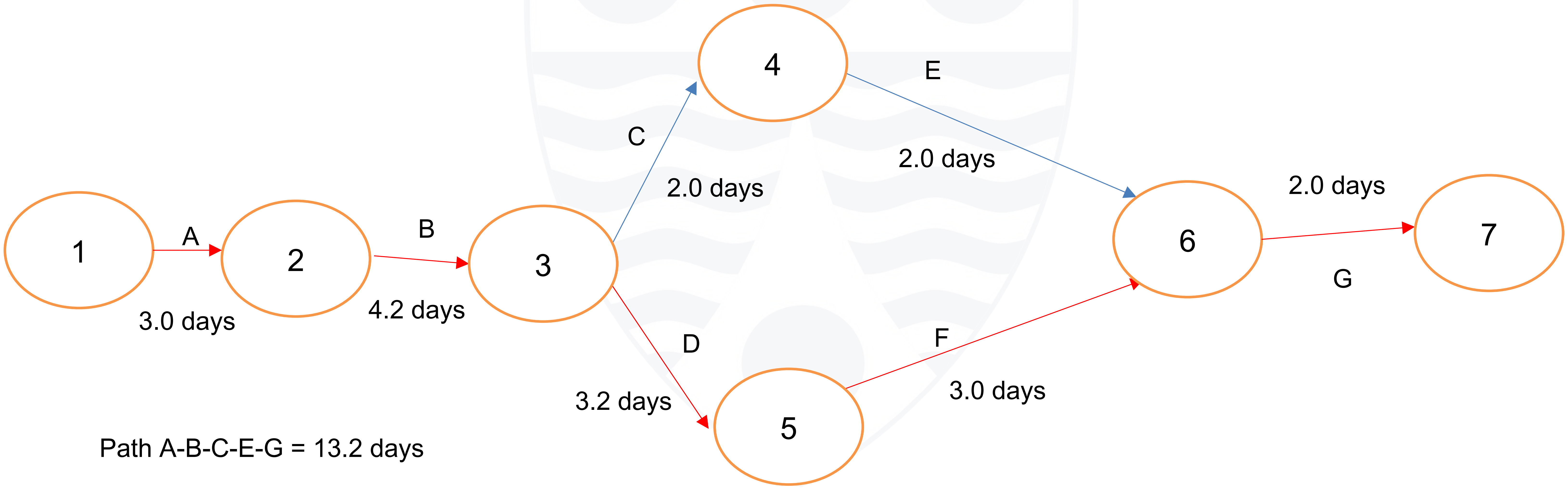
- You may plot the project's critical path on a Gantt Chart to more easily visualize the project's timelines. Both PERT and Gantt charts can be created on some scheduling software.

PERT [Practical Example]

Let us consider the same previous project to buy and install computers and solar system for a school. Assuming the following figures for optimistic, most likely and pessimistic time durations, we can compute the expected times (E) for each activity as follows:

Task Designation	Task Description	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected time (E)
A	Make a budget	2	3	4	3.0
B	Purchase items	3	4	6	4.2
C	Install computers	1	2	3	2.0
D	Install solar system	2	3	5	3.2
E	Test computers	1	2	3	2.0
F	Test solar system	2	3	4	3.0
G	Deployment	1	2	3	2.0

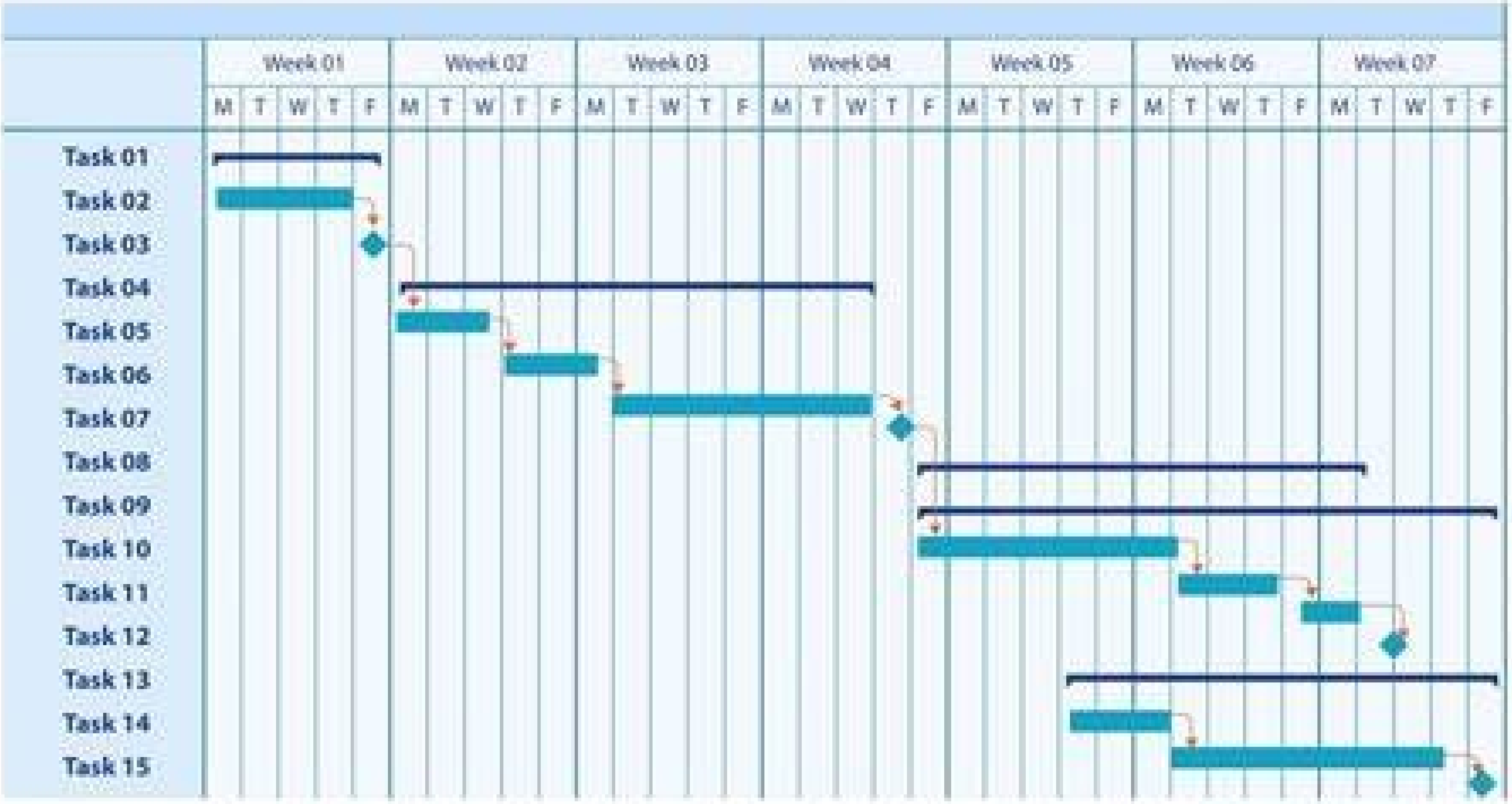
PERT [Solution to Problem]



Path A-B-C-E-G = 13.2 days

Path A-B-D-F-G = 15.4 days (Critical path)

Project Schedule on Gantt Chart



Control Schedule

Control schedule is the process of monitoring the status of the project to update the project schedule and managing changes to the schedule baseline.

Key Outputs

- **Work performance information** – this includes information on how the project work is performing compared to the schedule baseline.
- **Schedule forecasts** – these are forecasts of estimates or predictions of conditions and events in the project's future based on information and knowledge available at the time of the forecast.
- **Change requests** – Schedule variance analysis, as well as reviews of project reports, results of performance measures and modifications to the project scope or project schedule, may result in a change request to the schedule baseline, scope baseline or other components of the project management plan.

Control Schedule

Inputs	Tools & Techniques	Outputs
1. Project management plan -Schedule management plan -Schedule baseline -Scope baseline -Performance measurement baseline 2. Project documents -Lessons learned register -Project schedule -Project calendars -Resource calendars -Schedule data 3. Enterprise Environmental factors 4. Organizational process assets	1. Data analysis -Earned value analysis -Iteration burndown chart -Performance reviews -Trend analysis -Variance analysis -What-if scenario analysis 2. Critical path method 3. Schedule compression 4. Project management information system 5. Leads and lags 6. Resource optimization	1. Work performance information 2. Schedule forecasts 3. Change request 4. Project management plan updates -Schedule management plan -Schedule baseline -Cost baseline -Performance management baseline 5. Project documents updates -Assumption log -Basis of estimates -Lessons learned register -Project schedule -Resource calendars -Risk register

Common Problems in Project Scheduling

- Scope creep - unplanned changes in scope
- Resource management issues - inadequate resource allocations
- Tracking tasks and deadlines – may lead to missed deadlines
- Communication inadequacies – unclear or delayed communication
- Unrealistic deadlines – sometimes based on wrong assumptions in estimates

Control Measures

- Having a clear scope statement and implementing a change control process can help to reduce scope creep.
- Using project managing software can help in tracking project tasks and deadlines.
- Having a broad communication plan
- Schedule Compression techniques

Schedule Compression Techniques

Schedule compression techniques are used to shorten the schedule duration without reducing the project scope, in order to meet schedule constraints, imposed dates or other objectives.

- **Crashing** – a technique used to shorten the schedule duration for the least incremental cost by adding resources. Examples include:
  -approving overtime
  -bringing additional people and/or equipment
  -paying to expedite delivery to activities on the critical path

- **Fast tracking** – a technique in which activities or phases normally done sequentially are performed in parallel for at least a portion of their duration. Fast tracking only works when activities can be overlapped to shorten the project duration on the critical path.

Group Exercise and Presentation

- Choose any project of your choice with at least five activities, create a network diagram and show the critical path. There must be at least two paths in the network.

Conclusion

- Managing project schedule could be very demanding. Proper estimation of resources and durations is critical in planning schedules.
- Software tools can help in efficient schedule planning and tracking of activities and events.

Closing Remarks